

SIMATS ENGINEERING

Name	Register Number	Course Code	Course
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24. Perform Sharpening of Image using High-Boost Masks.

High-boost Masks

0	-1	0	-1	-1	-1
-1	$A+4$	-1	-1	$A+8$	-1
0	-1	0	-1	-1	-1

- $A \geq 1$
- if $A = 1$, it becomes "standard" Laplacian sharpening

AIM:

Sharpening of Image using High-Boost Masks.

PROGRAM:

```
import cv2
import numpy as np

def high_boost_filter(image, boost_factor):
    kernel_size = 3

    # Create averaging kernel
    kernel = np.ones((kernel_size, kernel_size), dtype=np.float32) / (kernel_size ** 2)

    # Blur the image
    blur_image = cv2.filter2D(image, -1, kernel)

    # Apply High-Boost Filtering
    mask = image + (image - blur_image) * boost_factor

    # Convert to 8-bit image
    mask = np.clip(mask, 0, 255).astype(np.uint8)

    return mask

image_path = r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png"

image = cv2.imread(image_path)

if image is None:
    print("Error: Image not found!")
else:
```

```
# Apply High-Boost Filter
sharpened_image = high_boost_filter(image, 1.5)

# Save the output image
cv2.imwrite(
    r"C:\Users\Susmitha\Downloads\ITA0510 Lab\sharpened_image.jpg",
    sharpened_image
)

# Display the images
cv2.imshow("Original Image", image)
cv2.imshow("High-Boost Sharpened Image", sharpened_image)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT:



OUTPUT:**RESULT:**

Thus the program is executed successfully.